EXPERIMENT NO. 8 ANALYSIS OF SERVER EVENT AND SECURITY LOGS

AIM:

To analyze Windows Security logs using Windows Event Viewer to identify security anomalies and investigate possible unauthorized activity by examining successful and failed login attempts, logoff events, and special privilege assignments.

OBJECTIVES:

1. To examine Windows Security logs using Windows Event Viewer and understand the security-related events recorded by the operating system.
2. To identify successful and failed login attempts and understand their importance in monitoring user authentication activities.
3. To analyze important security Event IDs and understand their meanings and significance in detecting potential security risks.
4. To examine logon sessions and special privilege assignments to identify unusual activities and understand how user access is managed.
5. To understand how security log analysis helps investigate possible unauthorized access, detect suspicious activities, and improve overall system security.

HARDWARE & SOFTWARE REQUIREMENTS:

Category	Specification & Tools
Hardware Configuration	PC/Laptop with Intel Core i5 or higher, minimum 4 GB RAM, Internet Connection
Operating System	Windows 10/11
Log Analysis Tool	Windows Event Viewer
System Utility	Run dialog (eventvwr.msc)
Security Log	Windows Logs → Security
Analysis Tools	Filter Current Log and Event Properties
Documentation Tools	Microsoft Word

THEORY:

Security log analysis is the process of examining system-generated records to identify unusual activities, security violations, and possible unauthorized access. Operating systems record events such as successful logins, failed authentication attempts, logoff activity, and privilege assignments. These records help administrators understand system activity and investigate potential security incidents.

Windows Event Viewer is a built-in Windows utility used to view and analyze system, application, and security events. It can be opened by executing eventvwr.msc in the Run dialog. The Security log contains information about authentication attempts, account management, and other security-related activities, depending on the auditing policies enabled on the system.

Windows security events are identified using unique Event IDs. Event ID 4624 represents a successful logon, while Event ID 4625 represents a failed logon attempt. Event ID 4634 indicates that a logon session ended, and Event ID 4672 indicates that special privileges were assigned to a logon. These events help users understand authentication activity and privilege-related operations.

Event Viewer allows users to filter logs using specific Event IDs and examine individual event properties. The General and Details tabs provide information about the recorded event, including its description, timestamp, and other available fields. This information can be used to investigate unusual login patterns and identify activities that require further examination.

Security log analysis helps identify repeated failed login attempts, unexpected successful logons, and unusual privilege assignments. However, a single event does not necessarily indicate an attack. The account details, timestamps, and related events must be examined before determining whether the activity is suspicious.

Windows Event Viewer provides a simple method for monitoring security-related activities and understanding the security status of a computer. Regular examination of these logs helps improve security awareness and supports the investigation of possible unauthorized access.

ANALYSIS OF WINDOWS SECURITY LOGS – PHASES, TOOLS & COMMANDS:

1. **Opening Windows Event Viewer**
   In this phase, Windows Event Viewer is opened to access operating system logs. The Security log is selected to examine authentication and security-related events recorded on the computer.

Tool / Utility	Purpose	Command Used
Run Dialog	Open Event Viewer	eventvwr.msc

Tool / Utility	Purpose	Command Used
Event Viewer	Access Windows logs	Windows Logs
Security Log	View security-related events	Windows Logs → Security

2. **Login Activity Analysis**

In this phase, successful and failed login attempts are examined using Event IDs 4624 and 4625. Event ID 4634 is also reviewed to understand when a logon session ends.

Tool / Utility	Purpose	Event ID Used
Windows Security Log	Examine successful logons	4624
Windows Security Log	Examine failed logons	4625
Windows Security Log	Examine ended logon sessions	4634

3. **Privilege Assignment Analysis**

In this phase, Event ID 4672 is examined to understand special privilege assignments associated with a logon. The event details are reviewed to identify activities that may require further investigation.

Tool / Utility	Purpose	Command Used
Windows Security Log	Examine special privilege assignments	4672
Event Properties	View recorded event information	General and Details tabs

4. **Filtering and Examining Security Events**

In this phase, relevant events are located using the Filter Current Log option. The General and Details tabs are examined to understand event descriptions, timestamps, and other available information.

Tool / Utility	Purpose	Command Used
Filter Current Log	Filter specific security events	Select Event IDs

Tool / Utility	Purpose	Command Used
Event Properties	Examine event descriptions	Double-click an event
Details Tab	Inspect additional event information	View event details

IMPLEMENTATION STEPS:

1. **Opening Windows Event Viewer**
   - Pressed Win + R to open the Run dialog.
   - Entered eventvwr.msc and pressed Enter to launch Windows Event Viewer.
   - Expanded Windows Logs in the left navigation panel.
   - Selected the Security log to view recorded security events.
   - Examined the available events and their timestamps.

2. **Login Activity Analysis**
   - Examined the Security log to identify successful and failed logon events.
   - Identified Event ID 4624 as a successful logon event.
   - Examined Event ID 4625 to understand failed logon attempts.
   - Reviewed Event ID 4634 to understand when logon sessions ended.
   - Examined the available event details to understand the recorded authentication activity.

3. **Privilege Assignment Analysis**
   - Examined Event ID 4672 in the Windows Security log.
   - Opened the event properties to inspect the available information.
   - Reviewed the event description to understand special privilege assignments.
   - Considered the account and logon details when interpreting the event.
   - Recorded the significance of privilege-related events for security monitoring.

4. **Filtering and Examining Security Events**
   - Selected the Security log in Windows Event Viewer.
   - Used the available event information to locate relevant security events.
   - Examined the Event IDs and their descriptions.
   - Opened individual events to view their General and Details tabs.
   - Reviewed the recorded information to understand the significance of the observed events.

OBSERVATIONS:

During the cybersecurity practical, Windows Event Viewer was used to examine Windows Security logs and analyze the security-related activities recorded by the operating system. The Security log was inspected to understand different types of authentication events, logon sessions, and privilege assignments. Important Event IDs were examined to understand their meaning and how they can help identify unusual activities. The recorded event information was reviewed to understand the role of security logs in monitoring system activity and investigating possible unauthorized access.

- Event ID 4672 was observed, indicating that special privileges were assigned to a logon.
- Event ID 4624 was observed, indicating successful logon activity.
- Event ID 4634 was observed, indicating that a logon session had ended.
- Event ID 4625 was observed, indicating a failed logon attempt.
- The event details provided information about the recorded security activities.
- The examined events demonstrated how Windows records authentication and privilege-related information.
- The observations showed how security logs can help investigate unusual login patterns and possible security anomalies.

OUTPUT:

Phase 1 – Opening Windows Event Viewer

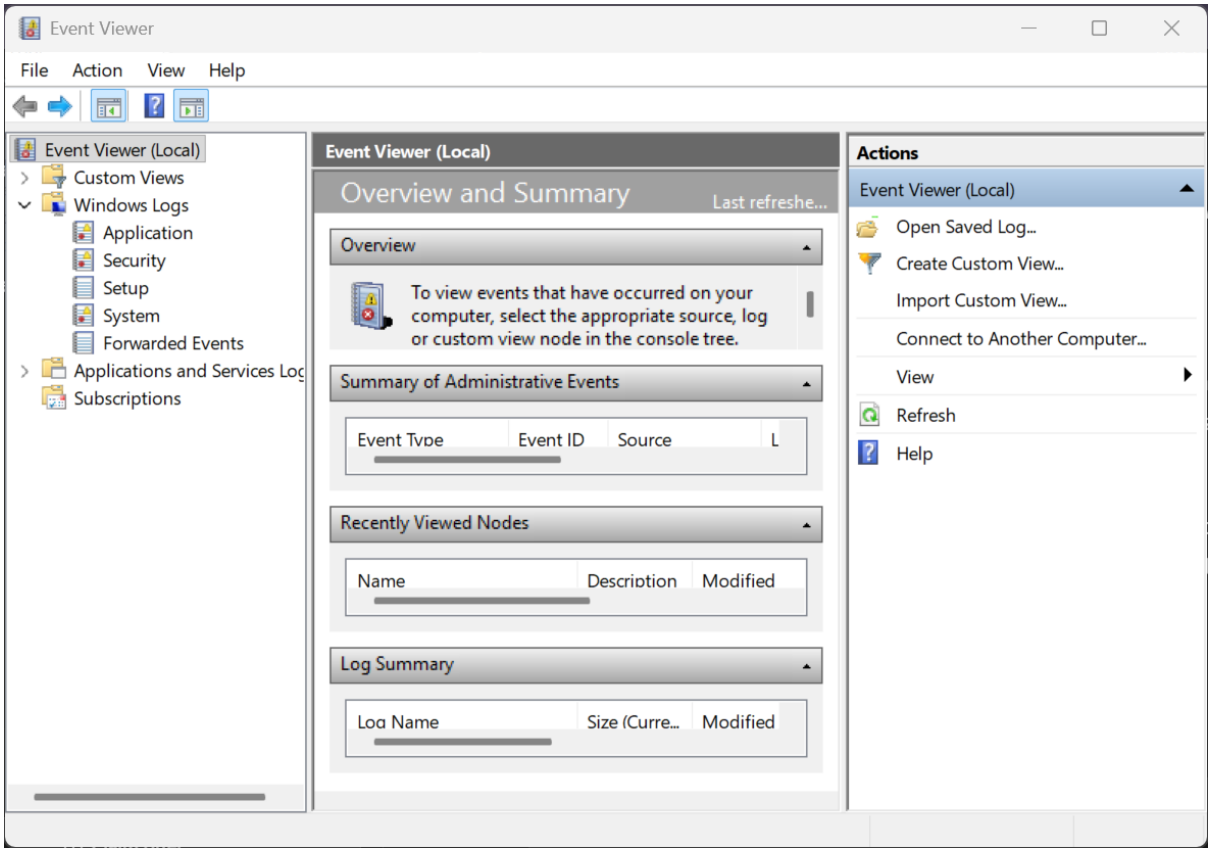


Fig. 1: Windows Event Viewer Overview and Summary

Phase 2 – Login Activity Analysis

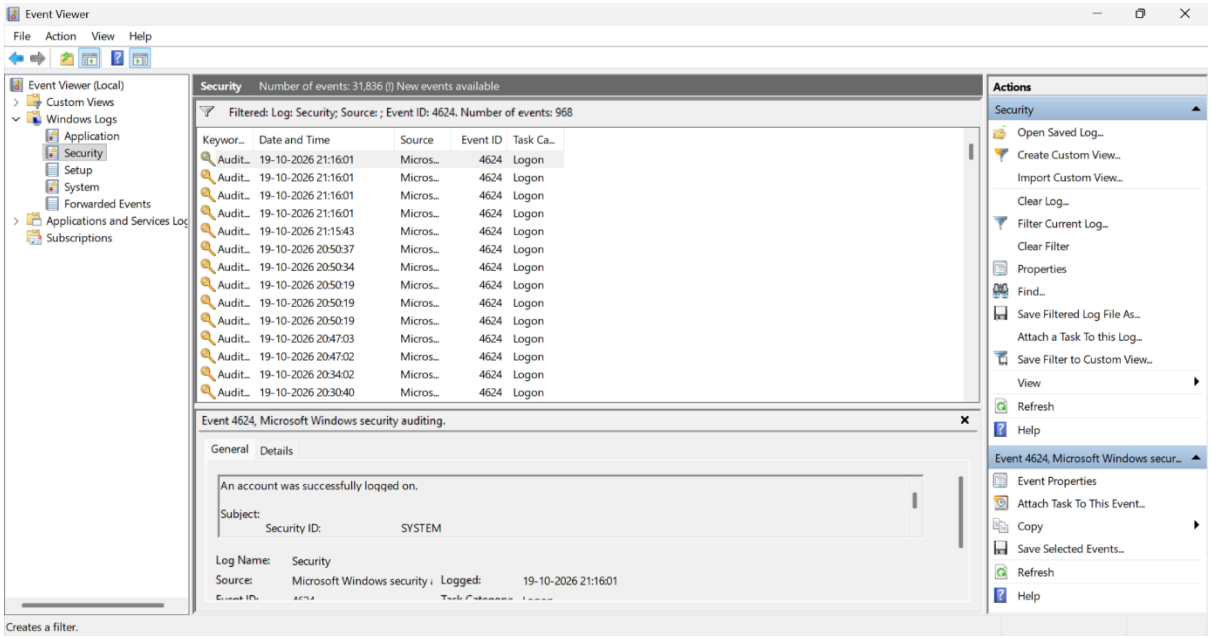


Fig. 2: Inspecting Successful Logon Activity (Event ID 4624)

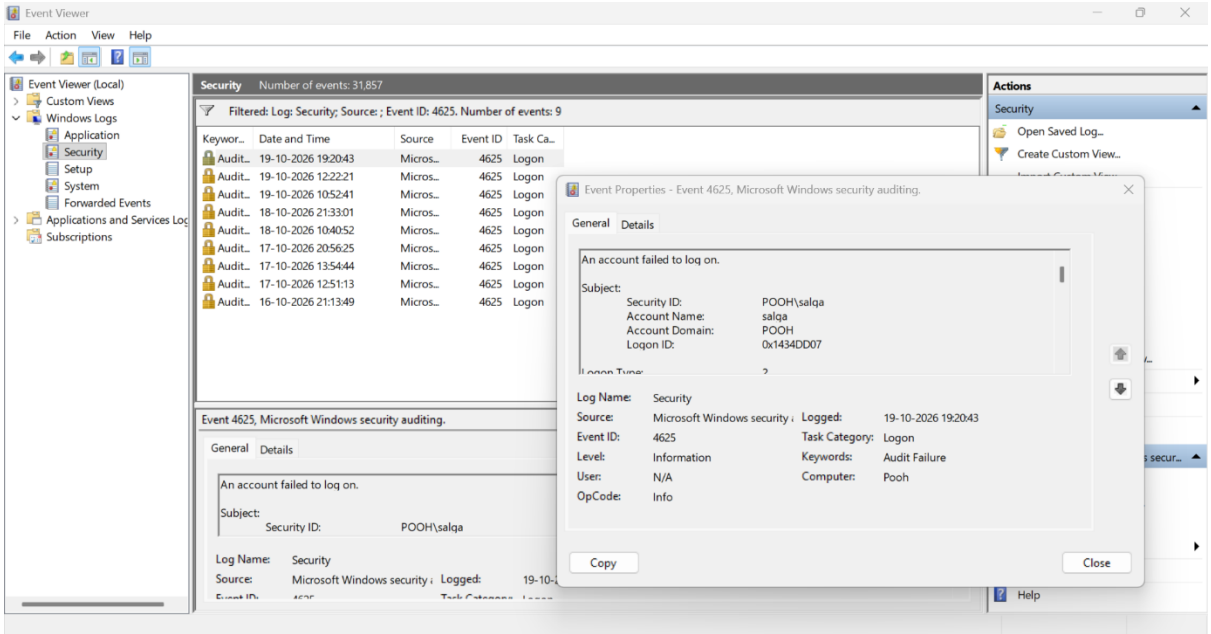


Fig. 3: Examining Failed Logon Events (Event ID 4625)

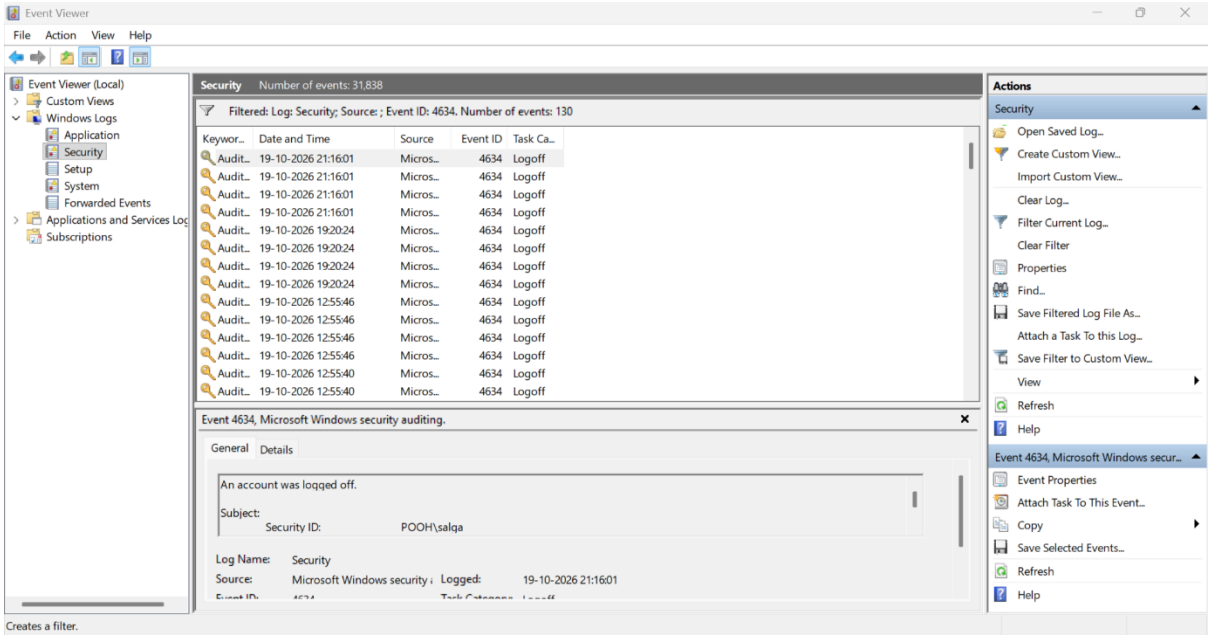


Fig. 4: Examining Logoff Events (Event ID 4634)

Phase 3 – Privilege Assignment Analysis

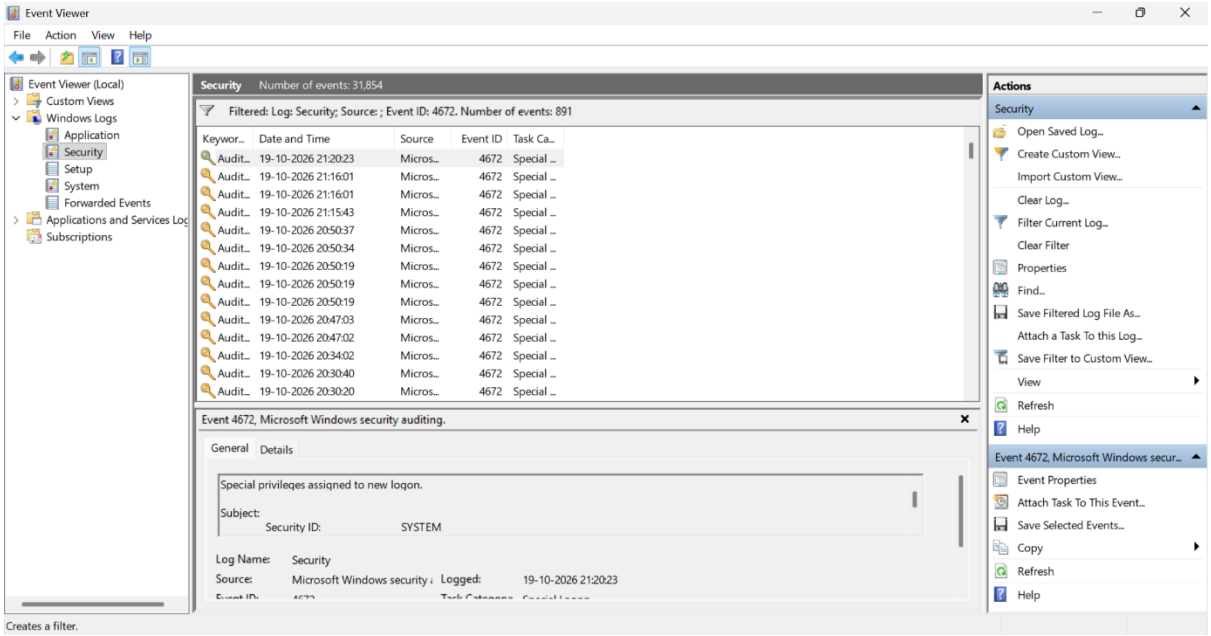


Fig.5: Examining Special Privilege Assignment Events

Phase 4 – Filtering and Examining Security Events

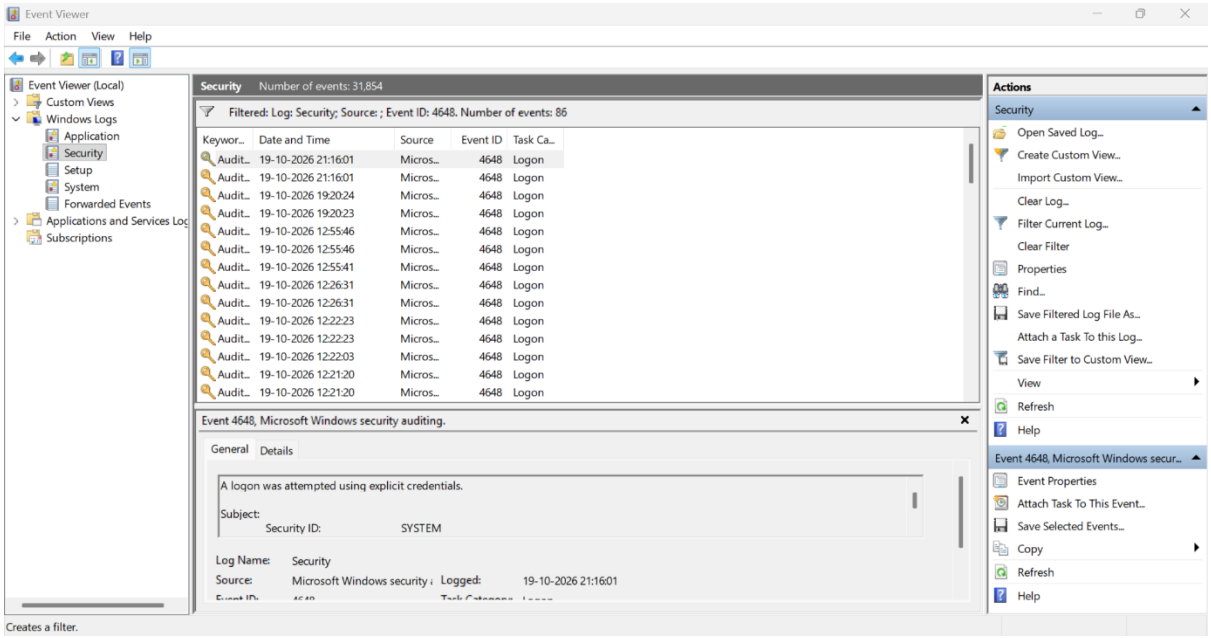


Fig. 4: Filtering and Examining Explicit Credential Logon Events (Event ID 4648)

ANALYSIS:

The experiment successfully demonstrated the analysis of Windows Security logs using Windows Event Viewer. Events related to successful logons, failed logons, ended sessions, and special privilege assignments were examined to understand their security significance. The observations showed how Windows records authentication activity and privilege-related operations. The practical also demonstrated how event properties can be used to investigate unusual activities. However, the observed events alone did not confirm unauthorized access, and further investigation of account details, timestamps, and related events would be required to establish whether a security incident had occurred.

PREVENTIVE MEASURES:

1. Keep the Windows operating system and installed applications updated with the latest security patches.
2. Use strong passwords and enable multifactor authentication wherever supported.
3. Monitor repeated failed login attempts to identify possible password-guessing activity.
4. Review user accounts and administrative privileges regularly to prevent unauthorized access.
5. Enable appropriate security auditing to maintain useful records of system activity.
6. Investigate unexpected successful logons and unusual privilege assignments.
7. Review security logs regularly to identify suspicious patterns and potential security risks.

CONCLUSION:

The experiment successfully demonstrated the analysis of Windows Security logs using Windows Event Viewer. Important events, including successful logons, failed logons, logoff activity, and special privilege assignments, were examined to understand their significance in security monitoring. The practical helped develop an understanding of how Windows logs can be used to investigate unusual activity and possible unauthorized access. It also highlighted the importance of regular security log monitoring in identifying potential security risks and maintaining system security.

.\accesschk.exe -d -v "$env:USERNAME" "C:\Users\$env:USERNAME"

- Examined the output to identify the access rights granted to the user on the specified directory.
- Recorded the observed permissions and reviewed the results as part of the Windows security audit.

OBSERVATIONS:

During the Windows security auditing practical, the Windows 11 Home Single Language system was examined using built-in Windows commands, PowerShell, gpresult, and Sysinternals AccessChk. The experiment helped evaluate system configuration, user privileges, execution policies, and directory permissions to understand the security settings of the computer.

- The system was identified as Windows 11 Home Single Language, Version 25H2, Build 26200.
- The whoami and net user commands displayed user identities, group memberships, and administrator accounts.
- The gpresult /r command showed user-level policies, while the computer-level audit encountered access denied.
- The PowerShell execution policy was RemoteSigned for the current user, with other scopes set to Undefined.
- secpol.msc was unavailable, while AccessChk reported RW and FILE_ALL_ACCESS for the user's profile directory.

OUTPUT:

Phase 1 – System Configuration Audit



Fig. 1: Windows System Configuration